

MAGNETOM

Replacement of Parts System

TFT Monitor - General

Document Version

Siemens reserves the right to change its products and services at any time.

In addition, manuals are subject to change without notice. The hardcopy documents correspond to the version at the time of system delivery and/or printout. Versions to hardcopy documentation are not automatically distributed.

Please contact your local Siemens office to order a current version or refer to our website <http://www.healthcare.siemens.com>.

Disclaimer

Siemens provides this documentation “as is” without the assumption of any liability under any theory of law.

The content described herein requires superior understanding of our equipment and may only be performed by qualified personnel who are specially trained for such installation and/or service.

Copyright

“© Siemens, 2016” refers to the copyright of a Siemens entity such as:

Siemens Healthcare GmbH - Germany

Siemens Aktiengesellschaft - Germany

Siemens Shenzhen Magnetic Resonance Ltd. - China

Siemens Shanghai Medical Equipment Ltd. - China

Siemens Healthcare Private Ltd. - India

Siemens Medical Solutions USA Inc. - USA

Siemens Healthcare Diagnostics Inc. - USA and/or

Siemens Healthcare Diagnostics Products GmbH - Germany

1	General Information	4
1.1	Safety Information	4
1.2	Monitor compatibility with workstation	5
2	Monitors	6
2.1	FRU Overview	6
2.2	Replacement Scenarios	7
2.2.1	Digital Workstation and Digital Monitor	7
2.2.2	Analog Workstation and Digital Monitor	7
2.2.3	Digital Workstation and Analog Monitor	8
3	TFT Color Monitor inspection	9
3.1	19" TFT Monitor - Visually inspecting/evaluating	9
3.1.1	Checking and preparing the Settings of the Graphics Card	9
3.1.2	Preparation for visual Evaluation for 19" TFT-Monitor	12
3.1.3	Visually checking the Geometry	13
3.1.4	Visually checking the Contrast	13
3.1.5	Checking the Luminance	14
3.2	24" TFT Monitor - Visually inspecting/evaluating	20
3.2.1	Checking and preparing the Settings of the Graphics Card	20
3.2.2	Calling up the test image	22
3.2.3	Visually checking the Geometry	24
3.2.4	Checking the Luminance	24
4	Changes to Previous Version	29
5	List of Hazard IDs	30

1.1 Safety Information

For your safety and the safety of others, as well as to protect the MR system and its components from damage, the appropriate safety information is included in the "General Safety Notes" and the "MR-Specific Safety Information" in the "Technical Documents."

Prior to starting work, read the specific information in the respective manual:

- General safety information for service personnel.
- Limitations with respect to ferromagnetic materials.
- Safety information pertaining to working with the magnet at field.
- Information regarding working with heavy loads.
- Protective conductor test before switching on the system and for subsequent repairs

 Adhere to ESD regulations and use the appropriate tool set: Anti-static wristband and grounded pad (TDK equipment type 8005 from 3M Deutschland GmbH).



CAUTION

Strong magnetic field!

Magnetic objects may cause injury and/or damage to sensitive parts.

» **Maintain a safe distance from the magnetic field.**



Protective ground connection test.

Measure the protective conductor resistance $\leq 100 \text{ m}\Omega$ with a ground test meter (Fluke 1503, or equivalent) between the protective conductor on the "sub-component" (e.g. RFPA) and the protective conductor connected to the "main component" (e.g. EPC).

Document the successful measurement in the service report (measured value and measuring equipment) as specified by IEC 62353.

1.2 Monitor compatibility with workstation

Tab. 1 Overview Monitor / Workstation

	Monitor	Material number	Workstation
Analog	6018-CC	3089083	R610
	SCD1897-M	3080991	R610
Digital 19"	DSC1904	8675402	R630 / R640 / R650
	DSC1907-D	10293002	R630 / R640 / R650
	DSC1908-D	10293050	R630 / R640 / R650
	DSC1910-D	10520199	R630 / R640 / R650 / SMC Avior / HP xw4600 / M470 / M470-2 / HP Z400 / HP Z420
	DSC1913-D	10656056	R630 / R640 / R650 / SMC Avior / HP xw4600 / M470 / M470-2 / HP Z400 / HP Z420
Digital 24"	MX241W ¹	10656011	M470 / M470-2 / Z400 / Z420 / Z440
	MX242W	11064630	Z440 / Z640 / Z240

1. Optional second monitor for syngo.via

2.1 FRU Overview

Tab. 2 FRU Overview

FRU	Material number	Could be replaced by	Final Checks	Instructions
6018-CC	3089083	3080991	Visually inspecting	
SCD1897-M	3080991	3080991	Visually inspecting	
DSC1904	8675402	10656056	Visually inspecting	(→ TFT Monitor DSC 1904-D / MR00-000.841.10)
DSC1907-D	10293002	10656056	Visually inspecting	(→ TFT Monitor DSC1907-D / MR00-000.841.07)
DSC1908-D	10293050	10656056	Visually inspecting	
DSC1910-D	10520199	10656056	Visually inspecting	(→ TFT Monitor DSC1910-D / MR00-000.841.08)
DSC1913-D	10656056	10656056	Visually inspecting	(→ TFT Monitor DSC1913-D / MR00-000.841.09)
MX241W	10656011	11064630	Visually inspecting	
MX242W	11064630	11064630	Visually inspecting	(→ Monitor MX242W / MR00-000.841.11)

2.2 Replacement Scenarios

2.2.1 Digital Workstation and Digital Monitor

2.2.1.1 Removal and Installation

1. Switch off the monitor.
2. Disconnect the cables from the old monitor at the host.
3. Remove the old monitor including cables.
4. Install the new monitor.
5. Connect the new cables, which are delivered with the monitor.

2.2.1.2 Final steps



Prior to evaluation, the TFT display must be in operation for at least 20 minutes (warm-up).

1. Switch on the monitor.
2. Allow the monitor to warm up (20 min), and then perform the following steps:
 - a) Visually inspect/evaluate the TFT monitor. (→ TFT Color Monitor inspection / Page 9)
3. If the visual inspection reveals any problems, refer to the document "Replacement of Parts" for the adjustment procedure of the respective TFT monitor.

2.2.2 Analog Workstation and Digital Monitor



Only for R610 Workstation.

2.2.2.1 Removal and Installation

1. Switch off the analog monitor.
2. Disconnect the cables from the old monitor at the host.
3. Remove the old monitor including cables.
4. Install the new digital monitor.
5. Remove the DVI → VGA Adapter on the R610 Workstation and connect the new cables delivered with the monitor directly to the DVI connector.



Some dedicated options (e.g. remote monitor) will not be functional in the future. They have to be replaced with the current version of the option.

2.2.2.2 Final steps



Prior to evaluation, the TFT display must be in operation for at least 20 minutes (warm-up).

1. Switch on the monitor.
2. Allow the monitor to warm up (20 min), and then perform the following steps:
 - a) Visually inspect/evaluate the TFT monitor. (→ TFT Color Monitor inspection / Page 9)
3. If the visual inspection reveals any problems, refer to the document “Replacement of Parts” for the adjustment procedure of the respective TFT monitor.

2.2.3 Digital Workstation and Analog Monitor



The old analog monitor must be replaced by a new digital monitor.

2.2.3.1 Removal and Installation

1. Switch off the monitor.
2. Disconnect the cables from the old monitor at the host.
3. Remove the old monitor including the cables.
4. Install the new monitor.
5. Connect the new cables, which are delivered with the monitor.

2.2.3.2 Final steps



Prior to evaluation, the TFT display must be in operation for at least 20 minutes (warm-up).

1. Switch on the monitor.
2. Allow the monitor to warm up (20 min), and then perform the following steps:
 - a) Visually inspect/evaluate the TFT monitor. (→ TFT Color Monitor inspection / Page 9)
3. If the visual inspection reveals any problems, refer to the document “Replacement of Parts” for the adjustment procedure of the respective TFT monitor.

3.1 19" TFT Monitor - Visually inspecting/evaluating

Prior to evaluation, the TFT display must be in operation for at least 20 minutes (warm-up).

3.1.1 Checking and preparing the Settings of the Graphics Card

3.1.1.1 Windows 7

- For settings of the graphics card:
 - Simultaneously press buttons **Ctrl** and **ESC** - the Start button appears
 - Click: **Start -> Control Panel -> Appearance and Personalization -> Display -> Change Display settings**
 - Change the appearance of your displays as follows:



The magnet display (Monitor 3) is not present on all system types.



The display settings below are an example only.
Monitor 2 can have the same resolution as the monitor 1.

Fig. 1: Display settings - main display

- Confirm by clicking OK

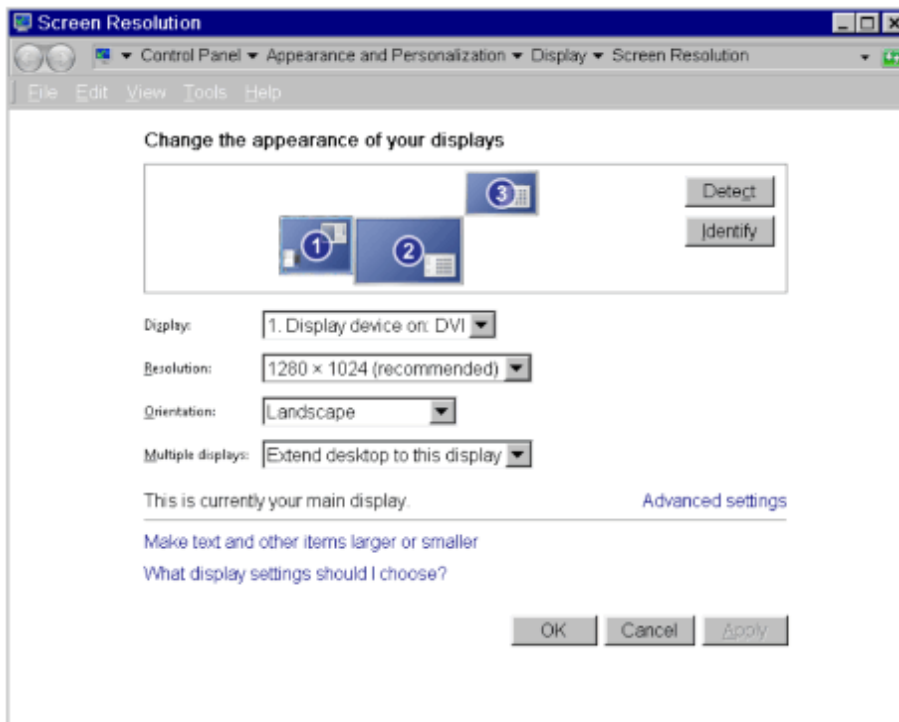
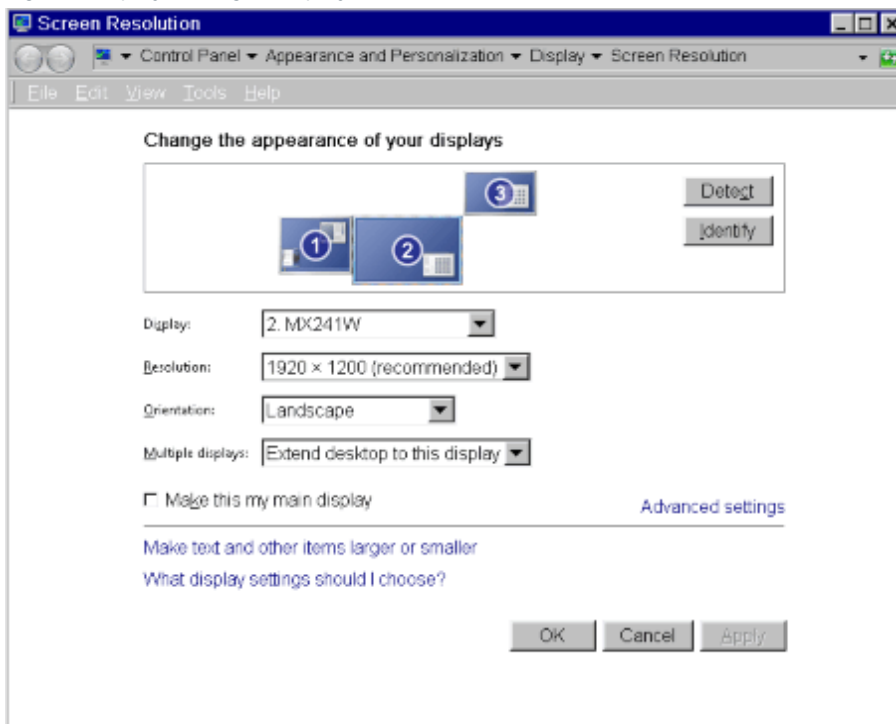
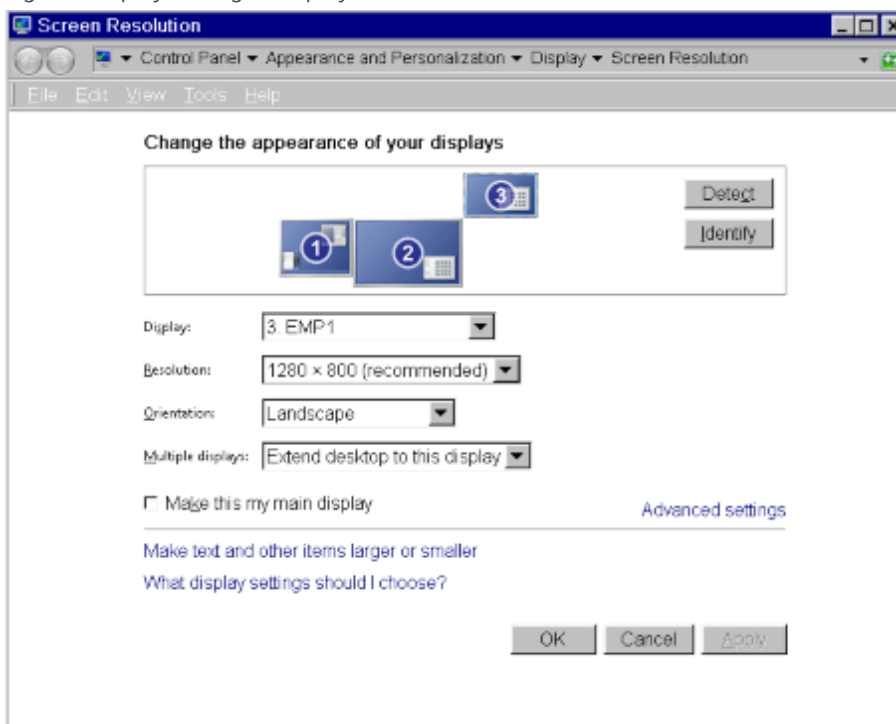


Fig. 2: Display settings - Display 2



- Confirm by clicking OK

Fig. 3: Display settings - Display 3



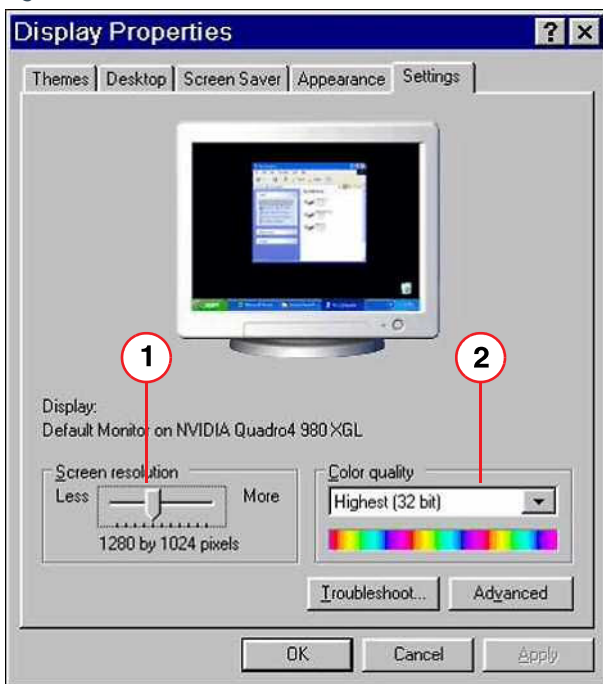
- Confirm by clicking OK

3.1.1.2 Windows XP

- Enable the Advanced User functionality:
 - Simultaneously press buttons **Ctrl** and **ESC** - the Start button appears

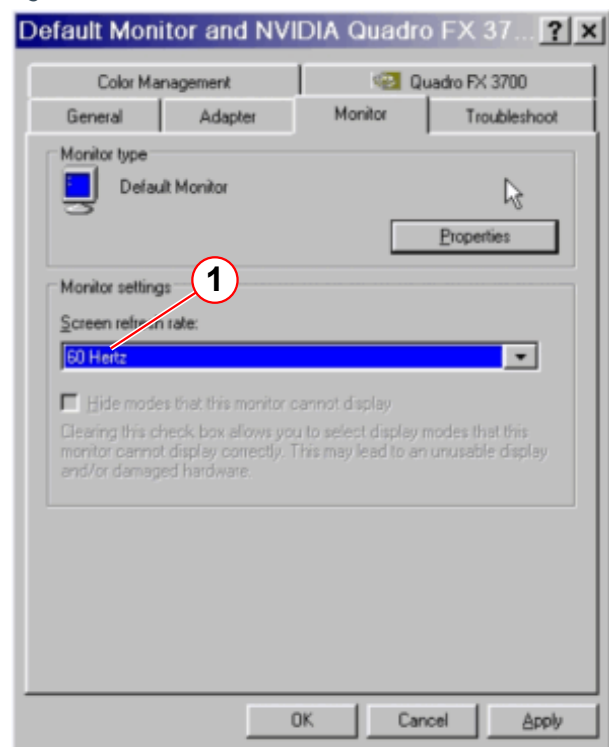
- Click: **Start** -> **Programs** -> **Advanced User** -> enter **PW** -> press **OK**
- The graphics card operates in **VESA SXGA / 1280 x 1024** mode at **60 Hz**.
- Display the Windows XP screen as follows:
 - Simultaneously press **Ctrl + Alt + Del**
 - Select the **Task Manager**
 - **Right-click** the **CapGM-Exam** in the list
 - In the new pop-up window click **Minimize** (WIN XP Desktop is now displayed in foreground)
- Check the **VESA mode** and **screen refresh rate**:
 - Right-click the background -> select **Properties**
 - Click the **Settings** tab card in the pop-up window
 - Check **VESA mode 1280 x 1024** (→ Fig. 4 Page 11)
 - Click the **advanced** button and select the **Monitor** tab card
 - Check the **screen refresh rate of 60 Hz** for TFTs(→ Fig. 5 Page 11)
 - Correct the setting, if necessary
 - Confirm each pop-up window by clicking **OK**

Fig. 4: VESA mode



- (1) 1280 x 1024
(2) True Color

Fig. 5: Screen refresh rate



- (1) 60 Hz.



The type of pop-up window depends on the type of graphics card used. It can differ from the examples (→ Fig. 4 Page 11) and (→ Fig. 5 Page 11)

3.1.2 Preparation for visual Evaluation for 19" TFT-Monitor



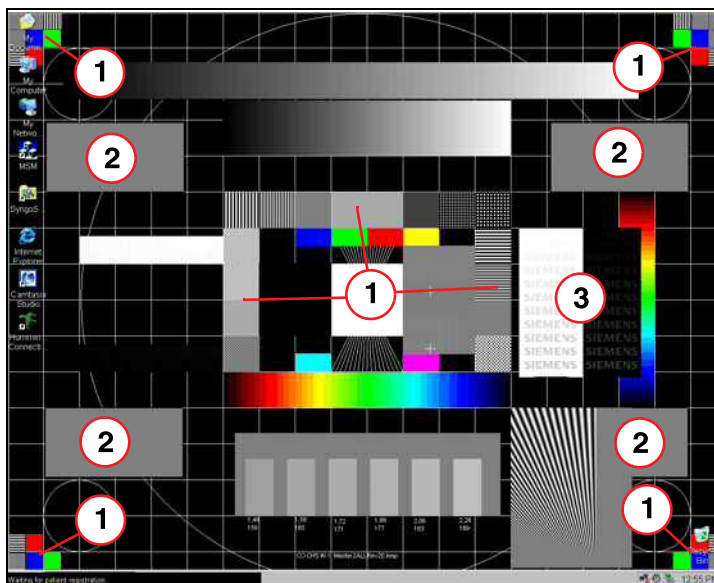
The display has to be in operation for at least 20 minutes.



The ambient light and monitor position must correspond to the operating conditions (to evaluate the criteria for contrast and patient images on-screen, the ambient light has to be at a dark level) and must be reproducible (this should be specified together with the customer).

- If the NUMARIS application is displayed again, minimize it as described above (task manager); result: the test image is displayed in full-screen mode (→ Fig. 6 Page 12).
- Select **test image Master2ALLRev20.bmp** as follows:
- **Right-click** the desktop background.
- Click **Properties** and select the **Desktop** tab card.
- Click **Browse**.
- Select the path:
C:\MedCom\utils\Imagequality\Master2AllRev20.bmp
- Click **Open** and **OK**, the result shows (→ Fig. 6 Page 12).

Fig. 6: Test image Master 2 All Rev 20



- (1) High frequency line pattern groups
- (2) Pixel on/off fields
- (3) SIEMENS logos (black and white)



Before starting to evaluate the monitors with regard to the image quality criteria, the Auto adjust program has to be run as described in the "**Adjustment procedure**" in the "Replacement of Parts" manual of the respective TFT monitor.

3.1.3 Visually checking the Geometry

3.1.3.1 Evaluation Criteria for the Geometry

- The image should be centered and fill the display area up to the edge; i.e., the image should be equidistant to all edges of the usable screen area.
- All screen information has to be visible along the edges of the displayed image; i.e., the double white frame must be visible on all sides of the screen.
- The horizontal/vertical ratio should be correct (squares have to appear square, circles are circles).
- Grid lines and other fields may not be distorted.
- If one or more of these criteria are not met, refer to the "Replacement of Parts" manual of the respective adjustment procedure for the TFT monitor.

3.1.4 Visually checking the Contrast

3.1.4.1 Evaluation Criteria for Contrast Setting

IMPORTANT:

- All 10 'SIEMENS' logos in the black and white fields (→ Fig. 6 Page 12) must be visible. The ambient light should correspond to the actual operating conditions on-site (this should be reproduced in consultation with the customer).
- The gradations in the upper gray scale should not be seen as steps.
- If one or more of these **criteria are not met**, proceed as per the directions "**Adjustment procedure**" in the "Replacement of Parts" manual for the respective TFT monitor.

3.1.5.2 Preparing the SMFitAct

Fig. 8: SMFit Act with diffusion screen



(1) Membrane switches



Each time the SMFitAct is switched on, it must be set to zero.

3.1.5.3 Zero Adjustment

The tube extension must be removed (unscrewed) for **direct measurement**.

- Press the **on/off** button, the display reading should be stable and not flashing.
- The letter '**D**' should be shown in the upper left corner of the display (meaning: **direct** measurement). If an '**A**' is displayed instead, press the **on/off** button again.
- Cover the SMFitAct tube completely, for example set it on a level surface (table) with the diffusion screen attached (→ Fig. 8 Page 15)
- Press both the **on/off** and **run/stop** switches at the same time.
- The value **0.00** should be displayed.

Additional instructions for operation can be found in the accompanying manual.

3.1.5.4 Measurement Procedure



When performing contact measurements using the SMFit Act spotmeter (with diffusion screen attached), ensure that you do not exert any pressure on the display panel while using the sensor. Otherwise, panel damage or malfunction may result.

3.1.5.5 Measurement of the white field (100% value)

- Without applying any pressure, center the diffusion screen in the 100% field (→ Fig. 9 Page 16). The **center hole** should be **filled completely** by the **100%** field. Carefully slide the SMFitAct probe through the center hole of the diffusion screen (→ Fig. 10 Page 17)
- Hold the spotmeter **exactly at right angles** to the screen, otherwise the measurement **values are not reliable**.

Read the 100% value and enter the data in (→ Tab. 3 Page 19) (100% - measured value - preadj.). The following measurement tolerances are permitted:

Recheck: 160 cd/m² (+/-7) cd/m²

Adjustment 160 cd/m² (+/-3) cd/m²



If setting values have been changed using the OSD menu, wait until the reading of the TFT display stops fluctuating (typically 5 to 10 seconds). The adjustment requires a certain amount of time to reach the desired state.

Fig. 9: Diffusion screen positioning at 100%

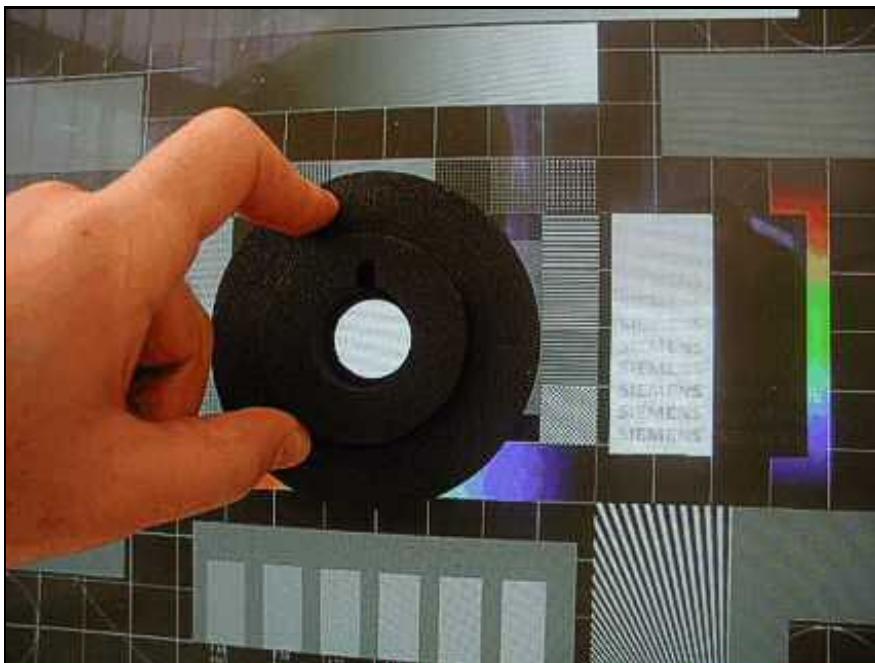
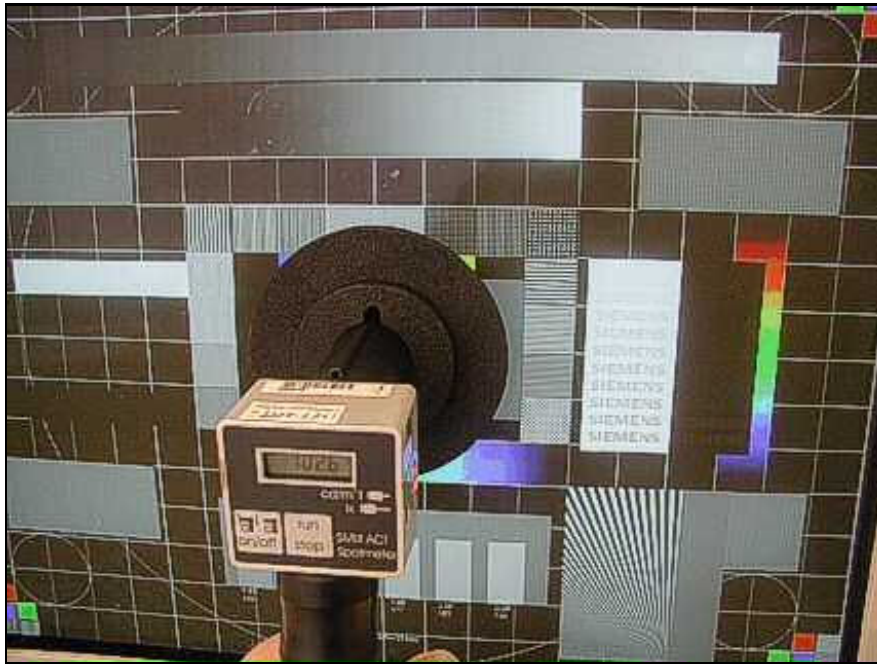


Fig. 10: Measurement position for 100%



3.1.5.6 Measurement of the black field (0% value)

- Proceed as with the 100% field, but use a different measurement position (→ Fig. 11 Page 18) and (→ Fig. 12 Page 18). The following measurement tolerances are permitted:

Recheck: $0.45 \text{ cd/m}^2 (+/-0.1) \text{ cd/m}^2$

Adjustment $0.45 \text{ cd/m}^2 (+/- 0.05) \text{ cd/m}^2$

- Enter the measured value in (→ Tab. 3 Page 19) under: 0% - measured value - preadj.

Fig. 11: Diffusion screen positioning at 0%

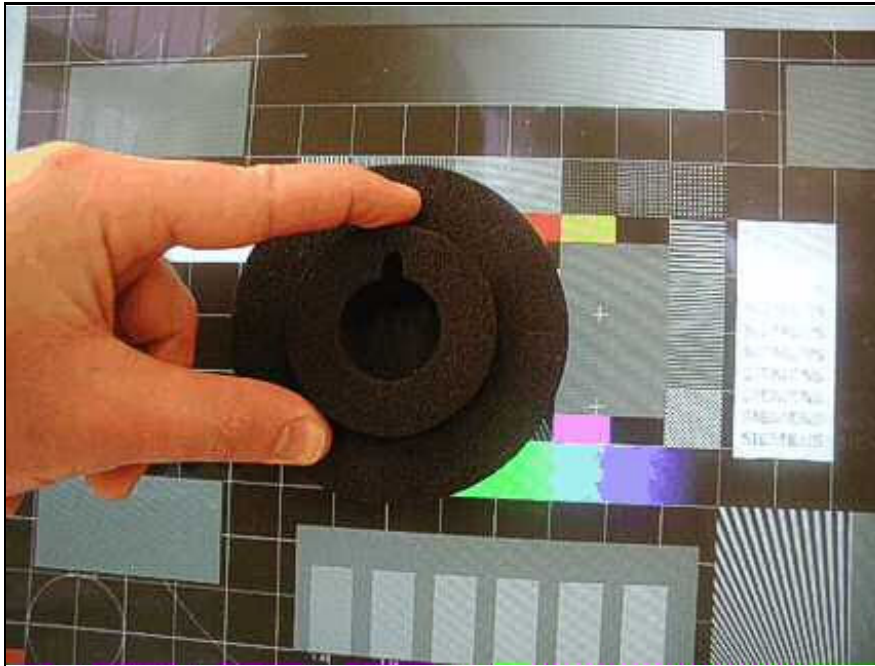


Fig. 12: Measurement position for 0%



If these tolerances (adjustment of the values of 0% and 100%) and the test criteria described above have been met, the monitor has already been adjusted to the corresponding PC. There is therefore no need for adjustment.

Tab. 3 Contrast, Brightness, Backlight - 19"

Monitor type: Serial no.:	MRC				MRSC			
	Measured value		Slider position		Measured value		Slider position	
	Preadj.	Correct.	Preadj.	Correct.	Preadj.	Correct.	Preadj.	Correct.
100% /Contrast								
0% /Brightness								
Backlight:	-----	-----			-----	-----		
Frequency	-----	-----			-----	-----		
Phase/Phase shift	-----	-----			-----	-----		

- If one or both measured values are not within tolerance, proceed with the TFT monitor as described in the "**Adjustment procedure**" of the "Replacement of Parts" manual for the respective TFT monitor.



Reset the screen background to normal

- Click the **right mouse button** once, a window pops up.
- Click the menu item **Properties**.
- Choose the **Desktop** tab card.
- Select **None** from the list and confirm with **OK**.

3.2 24" TFT Monitor - Visually inspecting/evaluating

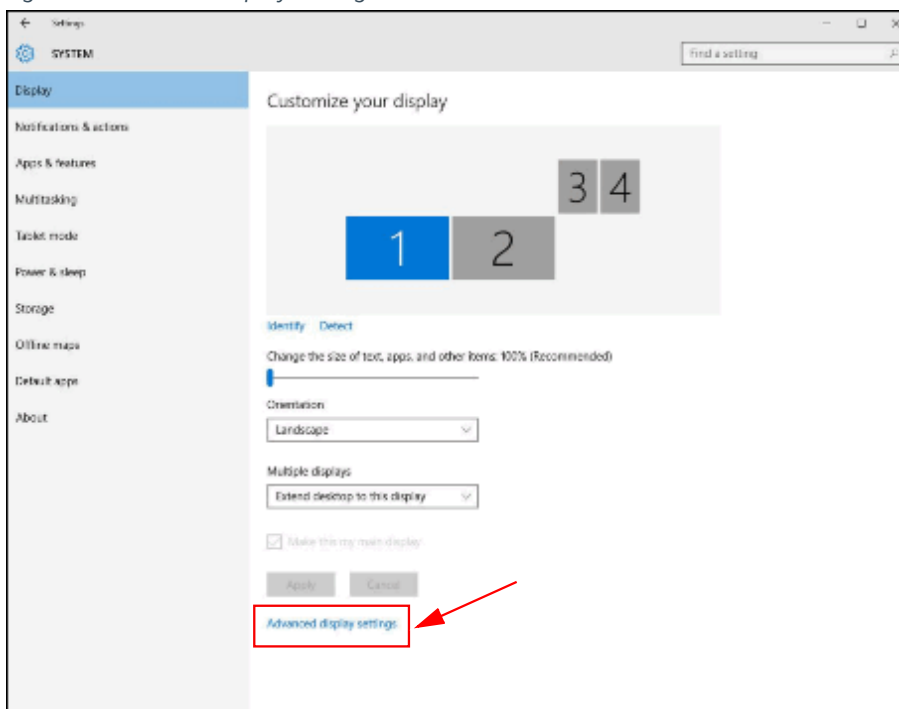
Prior to evaluation, the TFT display must be in operation for at least 20 minutes (warm-up).

3.2.1 Checking and preparing the Settings of the Graphics Card

3.2.1.1 Windows 10

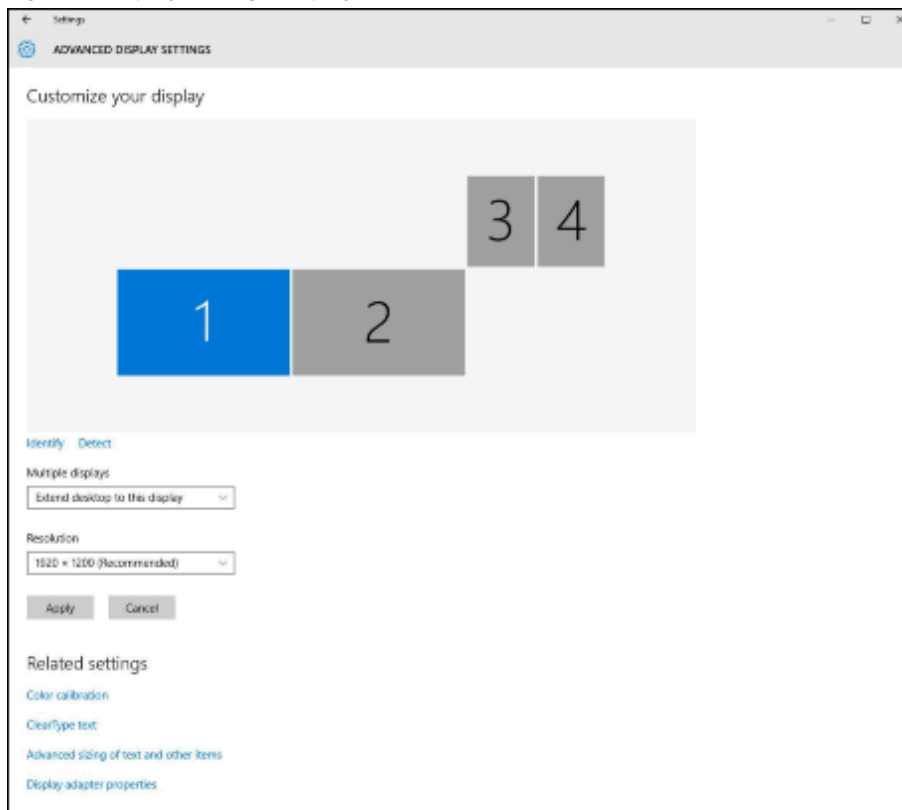
- To leave the kiosk mode:
 - Simultaneously press **TAB** and **DEL** and **↵** (send to node 1) keys on the console.
 - Enter user and password
- For settings of the graphics card:
 - Simultaneously press buttons **Ctrl** and **ESC** - the Start button appears
 - Click: **Start** -> **Setting** -> **System** -> **Display** -> **Customize your display**

Fig. 13: Advanced Display settings



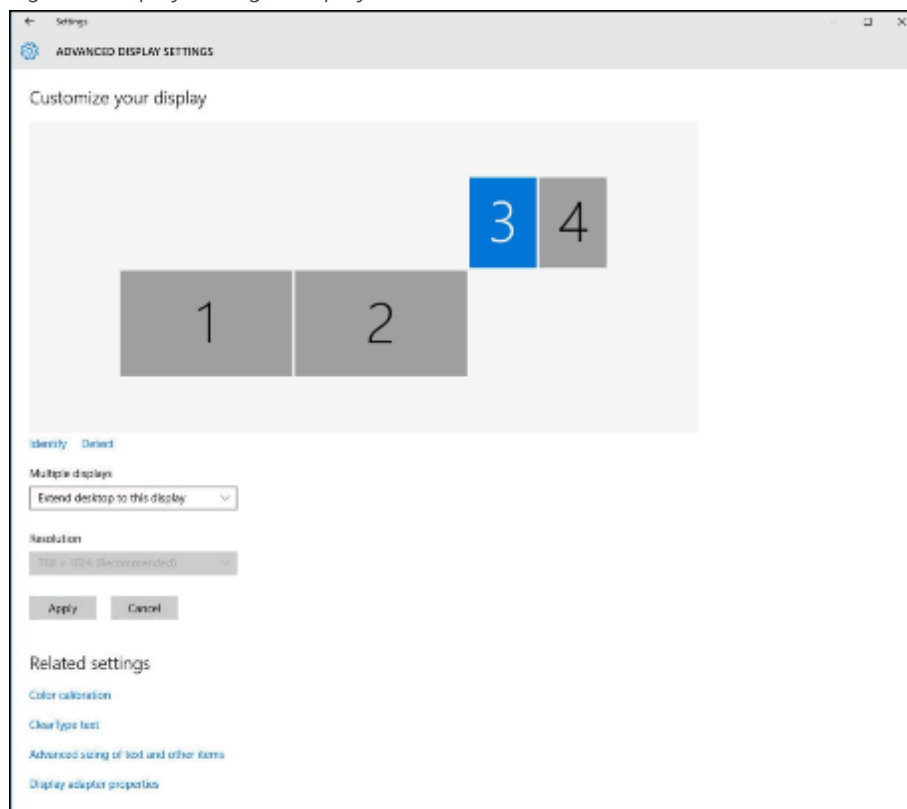
- Open **Advanced display settings**

Fig. 14: Display setting - Display 1 and 2



- Change the Resolution, if necessary to the recommended rate.

Fig. 15: Display settings - Display 3 and 4



- The resolution for Display 3 and 4 can not be changed.

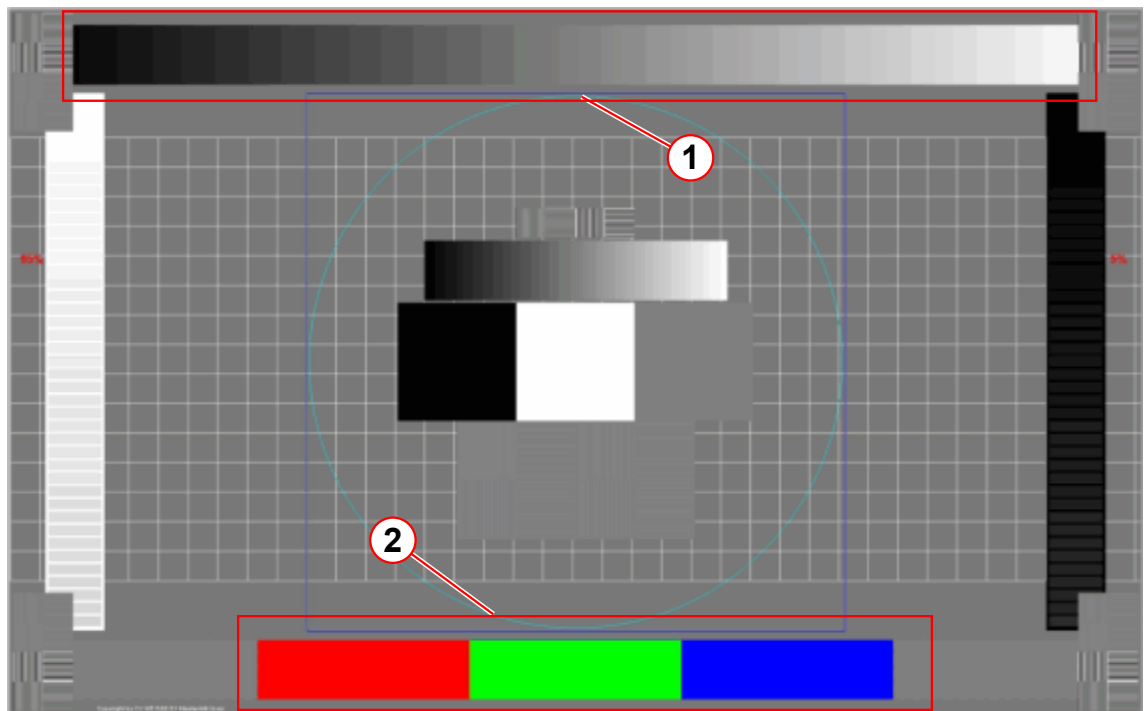
3.2.2 Calling up the test image



The display has to be in operation for at least 20 minutes.

- Simultaneously press **TAB** and **DEL** and **[→]** (send to node 1) keys on the console to leave the kiosk mode.
 - » Enter user and password
- Open the file explorer (by pressing **Windows** and the **E** key).
- Navigate to **This PC > K:\ > SeSo > TestImages**
- Double-click the file name: **Master4All_1920x1200.bmp**
 - » Paint opens with the test image.
- Move the **Paint** window over to the left monitor.
- Open the test image in full screen mode (press key **F11** or select tab card **View > Full screen**).

Fig. 16: Test image - Master4All_1920x1200



- (1) Gray-scale check
- (2) Color check

3.2.2.1 Checking the TFT monitor

If any adjustment is necessary, refer to 'Replacement of Parts REP'; 'Visually inspecting/ evaluating the TFT monitor' in detail.

- Check the MRAWP (MRC) TFT monitor using the test image.(→ Fig. 16 Page 23)
 - » All colors are displayed.
 - » Gray-scale is correctly displayed.
 - » Geometry is correct.
- Press the **ESC** key to close the full screen mode.
- Move the **Paint** window over to the right monitor.
- Open the test image in full screen mode (press key **F11** or select tab card **View > Full screen**).
- Check the MRAWP TFT monitor using the test image. (→ Fig. 16 Page 23)
 - » All colors are displayed.
 - » Gray-scale is correctly displayed.
 - » Geometry is correct.
- If the MRWP (MRSC) TFT monitor (option) installed, check the MRWP TFT monitor following the same procedure.

3.2.2.2 Set the normal mode.

- Press the **ESC** key to close the full screen mode.

- Close Paint and the file explorer.

3.2.3 Visually checking the Geometry

3.2.3.1 Evaluation Criteria for the Geometry

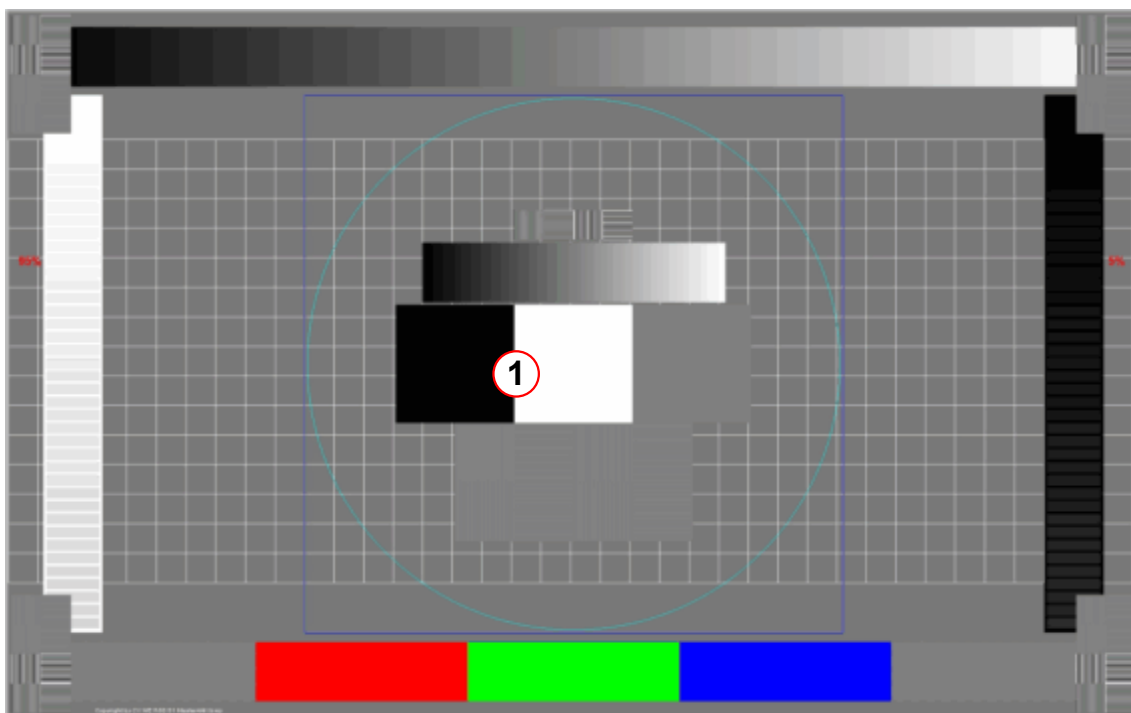
- The image should be centered and fill the display area up to the edge; i.e., the image should be equidistant to all edges of the usable screen area.
- All screen information has to be visible along the edges of the displayed image; i.e., the double white frame must be visible on all sides of the screen.
- The horizontal/vertical ratio should be correct (squares have to appear square, circles are circles).
- Grid lines and other fields may not be distorted.

3.2.4 Checking the Luminance

3.2.4.1 Preliminary step:

- The Siemens test image Master4All_1920x1200 is open as the desktop background.

Fig. 17: Test image - Master4All_1920x1200



(1) Measurement fields for 0% and 100% values

- The **white and black test fields** in the center of the test image are used to **measure** and **adjust** the image (marked in).(→ Fig. 17 Page 24)

3.2.4.2 Preparing the SMFitAct

Fig. 18: SMFit Act with diffusion screen



(1) Membrane switches



Each time the SMFitAct is switched on, it must be set to zero..

3.2.4.3 Zero Adjustment

The tube extension must be removed (unscrewed) for **direct measurement**.

- Press the **on/off** button, the display reading should be stable and not flashing.
- The letter '**D**' should be shown in the upper left corner of the display (meaning: **direct** measurement). If an '**A**' is displayed instead, press the **on/off** button again.
- Cover the SMFitAct tube completely, for example set it on a level surface (table) with the diffusion screen attached (→ Fig. 18 Page 25)
- Press both the **on/off** and **run/stop** switches at the same time.
- The value **0.00** should be displayed.

Additional instructions for operation can be found in the accompanying manual.

3.2.4.4 Measurement Procedure



When performing contact measurements using the SMFit Act spotmeter (with diffusion screen attached), ensure that you do not exert any pressure on the display panel while using the sensor. Otherwise, panel damage or malfunction may result.

3.2.4.5 Measurement of the white field (100% value)

- Without applying any pressure, center the diffusion screen in the 100% field (→ Fig. 19 Page 26). The **center hole** should be **filled completely** by the **100%** field. Carefully slide the SMFitAct probe through the center hole of the diffusion screen (→ Fig. 20 Page 27)
- Hold the spotmeter **exactly at right angles** to the screen, otherwise the measurement **values are not reliable**.

Read the 100% value and enter the data in (→ Tab. 4 Page 28) (100% - measured value - preadj.). The following measurement tolerances are permitted:

Recheck: 180 cd/m² (+/-7) cd/m²

Adjustment 180 cd/m² (+/-5) cd/m²



If setting values have been changed using the OSD menu, wait until the reading of the TFT display stops fluctuating (typically 5 to 10 seconds). The adjustment requires a certain amount of time to reach the desired state.

Fig. 19: Diffusion screen positioning at 100%

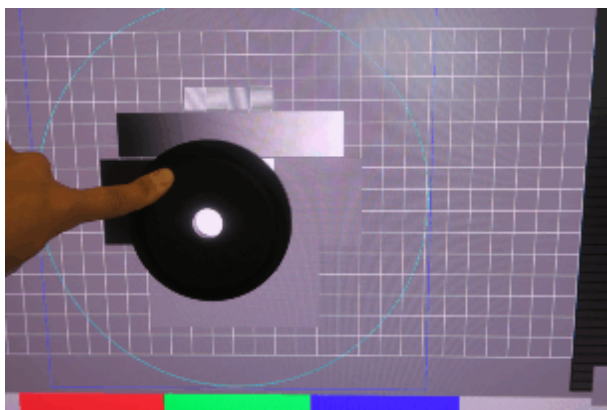
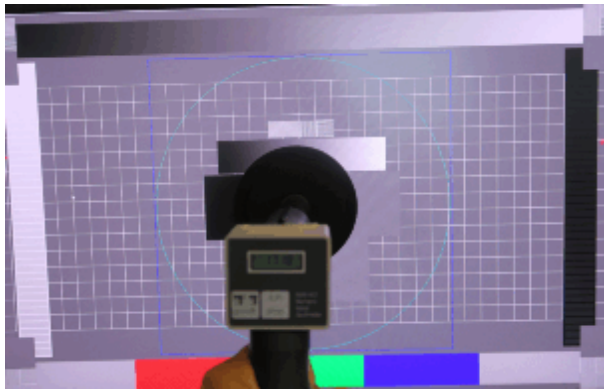


Fig. 20: Measurement position for 100%



3.2.4.6 Measurement of the black field (0% value)

- Proceed as with the 100% field, but use a different measurement position (→ Fig. 21 Page 27) and (→ Fig. 22 Page 27). The following measurement tolerances are permitted:

Recheck: $0.35 \text{ cd/m}^2 (+/-0.15) \text{ cd/m}^2$

Adjustment $0.35 \text{ cd/m}^2 (+/- 0.1) \text{ cd/m}^2$

- Enter the measured value in (→ Tab. 4 Page 28) under: 0% - measured value - preadj.

Fig. 21: Diffusion screen positioning at 0%

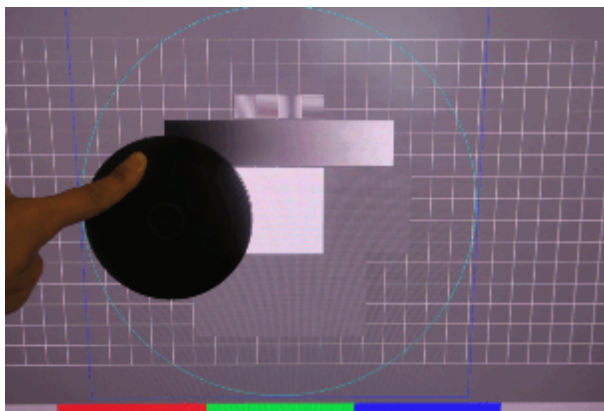
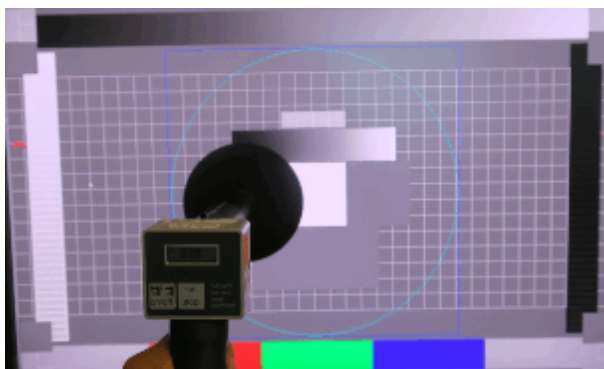


Fig. 22: Measurement position for 0%



If these tolerances (adjustment of the values of 0% and 100%) and the test criteria described above have been met, the monitor has already been adjusted to the corresponding PC. There is therefore no need for adjustment.

Tab. 4 Contrast, Brightness, Backlight - 24"

Monitor type: Serial no.:	MRAWP				MRWP			
	Measured value		Position		Measured value		Position	
	Preadj.	Correct.	Preadj.	Correct.	Preadj.	Correct.	Preadj.	Correct.
100% /Contrast								
0% /Brightness								
Backlight:	-----	-----			-----	-----		
Frequency	-----	-----			-----	-----		
Phase/Phase shift	-----	-----			-----	-----		

- If one or both measured values are not within tolerance, proceed with the TFT monitor as described in the "**Adjustment procedure**" of the "Replacement of Parts" manual for the respective TFT monitor.

Version 02

- Sempra was added.

Version 03

- Monitor MX242W for Vida and Visually inspecting/evaluating for 24 Zoll TFT Monitor were added.

There are no Hazard IDs in this document.

All documents may only be used for rendering services on Siemens Healthcare Products. Any document in electronic form may be printed once. Copy and distribution of electronic documents and hardcopies is prohibited. Offenders will be liable for damages. All other rights are reserved.

healthcare.siemens.com/services

Siemens Healthineers Headquarters
Siemens Healthcare GmbH
Henkestr. 127
91052 Erlangen
Germany
Telephone: +49 9131 84-0
siemens.com/healthineers

Print No.: MR00-000.841.06.03.02 | Replaces: MR00-000.841.06.02.02
Doc. Gen. Date: 06.17 | Language: English
© Siemens Healthcare GmbH, 2016

siemens.com/healthineers